

Extra — Nested Data: Searching a Real Pokédex

Topic: Nested Dictionaries + Lists of Dictionaries · **Course:** Extra Worksheet · **Time:** about 45 minutes

Keep these words handy: **nested**, **key**, **index**, **unpack**, **.get()**, **KeyError**

A real Pokédex file is not flat. A name gives you a dictionary. Inside it, **"stats"** gives you another dictionary, and **"moves"** gives you a **list** of dictionaries. Each step down is one more **[]**. This worksheet is about reading those steps on paper, then adding one layer to your own app.

This is the shape of the data your app imports:

```
# pokemon_info.py
pokedex = {
    "bulbasaur": {
        "id": 1,
        "types": ["grass", "poison"],
        "description": "A strange seed was planted on its back at birth.",
        "stats": {"hp": 45, "attack": 49, "defense": 49, "speed": 45},
        "habitat": "grassland",
        "moves": [
            {"name": "tackle", "level_learned": 1},
            {"name": "growl", "level_learned": 1},
            {"name": "leech seed", "level_learned": 7},
            {"name": "vine whip", "level_learned": 13},
            {"name": "poison powder", "level_learned": 20},
            {"name": "razor leaf", "level_learned": 27},
            {"name": "growth", "level_learned": 34},
            {"name": "sleep powder", "level_learned": 41},
            {"name": "solar beam", "level_learned": 48},
        ],
    },
}
```

1 · Predict 🌟

Read each line. Write what it prints — just from reading.

```
print(pokedex["bulbasaur"]["habitat"])
```

One key, then one more key. What prints?


```
print(pokedex["bulbasaur"]["types"][0])
```

"types" holds a list. What does [0] pick out?


```
print(pokedex["bulbasaur"]["moves"][2]["name"])
```

Three steps: key, index, key. What prints?


```
print(len(pokedex["bulbasaur"]["moves"]))
```

What number prints, and what is it counting?


```
for k in pokedex["bulbasaur"].keys():  
    print(k)
```

How many lines print? Write them all.

2 · Find the Path

Write the code that reaches each value. Use the data on page 1.

The **speed** stat of bulbasaur.

The level at which bulbasaur learns **vine whip** .

The second type in the **types** list.

The name of the last move in the list — without counting the moves yourself.

3 · Spot the Bug 🐛

Read what each block should do. Fix it on paper, then explain the fix.

Bug A — This should show move number 3. The user types `3`, but the wrong move appears.

```
number = int(input("which move? "))
print(pokedex["bulbasaur"]["moves"][number]["name"])
```

Hint: the user counts from 1. Python counts from 0.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should find bulbasaur. The user types `Bulbasaur` with a capital letter and a space, and the app says it is not a Pokémon.

```
search = input("what pokemon? ")
if search not in pokedex:
    print("not a pokemon")
```

Hint: clean the text before you search with it.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should print the level of the first move. It crashes.

```
print(pokedex["bulbasaur"]["moves"]["level_learned"])
```

Hint: `"moves"` is a list, not a dictionary. Which move do you mean?

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug D — The user types `type` instead of `types`. The app crashes with a `KeyError` instead of saying something helpful.

```
more_info = input("what info? ")  
print(pokedex["bulbasaur"][more_info])
```

Hint: `.get(key, default)` returns the default instead of crashing.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Explain the Code

This is the search app you built in your lesson. Read it, then answer from reading only.

```
import pokemon_info

search = input('what pokemon do you like?')

if search not in pokemon_info.pokedex:
    print(f'you searched {search}')
    print("not a pokemon. try again and check your spelling.")
else:
    print(f'you searched {search}')

    for pokemon_key in pokemon_info.pokedex[search].keys():
        print(pokemon_key)

    more_info = input('what piece of info do you want')

    if more_info == 'moves':
        print('you have a lot of moves. what info will you unpack?')
        print(f'The number of things to unpack is ...
{len(pokemon_info.pokedex[search][more_info])}')

        moves = int(input('what move do you want to unpack? [1-8]'))

        print(pokemon_info.pokedex[search][more_info][moves-1])

        moves_more_info = input('what are you going to know more about? the levels
learned or the name of that move?')

        if moves_more_info == 'name':
            print(pokemon_info.pokedex[search][more_info][moves-1]
[moves_more_info])
        elif moves_more_info == 'levels learned':
            print(pokemon_info.pokedex[search][more_info][moves-1]
['level_learned'])

    else:
        print(pokemon_info.pokedex[search][more_info])
```


What does the `if search not in pokemon_info.pokedex:` line protect the program from?

The `.keys()` loop runs before the second question. Why is that helpful to the user?

Explain `moves-1` to someone who has never coded. Why not just `moves` ?

The prompt says `[1-8]` , but bulbasaur has 9 moves. What can the user never reach?

If the user answers `stats` , the last `else` line runs. What exactly appears on the screen?

Where in this program could a `KeyError` still happen? Point at the line.

app.english-coding.co.uk and grow **main.py**

Bulbasaur and BULBASAUR

[1-8]

3. No crashes. If the user asks for a key that does not exist, print a friendly message instead of letting the program die.

Stretch: after showing one Pokémon, ask "search again?" and go back to the top instead of ending.

Write your plan here before you code.

6 · Explain Your Code 🎥

Record a short video on your phone explaining the code **you** wrote. Try to use these words: **nested**, **key**, **index**, **unpack**, **.get()**.

1. Show `pokemon_info.py` and point at one Pokémon's `moves` list.
2. Run your app and search for a Pokémon.
3. Walk through how one value travels from the data file to the screen.
4. Show the numbered move list and say why it is better than `[1-8]` .

Plan what you will say here before you record.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.